6 THICK TAILS IN HIGHER DIMENSIONS†

THIS DISCUSSION is about as simplified as possible handling of higher dimensions. We will look at 1) the simple effect of fat-tailedness for multiple random variables, 2) Ellipticality and distributions, 3) random matrices and the associated distribution of eigenvalues , 4) How we can look at covariance and correlations when moments don't exist (say, as in the Cauchy case).

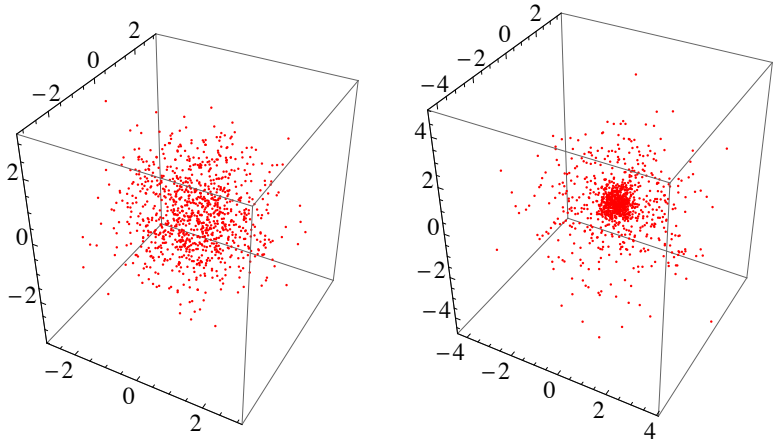

Figure 6.1: *Thick tails in higher dimensions: For a 3 dimentional vector, thin tails (left) and thick tails (right) of the same variance. In place of a bell curve with higher peak (the "tunnel") of the univariate case, we see an increased density of points towards the center.*

† Discussion chapter.